Enhanced Superconducting Transition Temperature in Electroplated Rhenium

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We show that electroplated Re films in multilayers with Cu, Au, and Pd show enhancement in superconducting critical temperature relative to previous methods of preparing Re. The dc resistance and magnetic susceptibility indicate a critical temperature approximately 6 K. Magnetization as a function of magnetic field at 1.8 K demonstrates type-II superconductivity, with an upper critical field on the order of 2 T. Critical current densities greater than 10^7 A/m^2 were measured above liquid-helium temperature. Low rf-loss was obtained below the critical temperature for multilayers deposited onto resonators made with Cu traces on commercial circuit boards. These superconducting films can be integrated into a wide range of standard components for low-temperature electronics.

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Versatile superconducting materials are important in order to continue the progress of high speed classical [1–4] and quantum computers[5] as well as a wide range of information technology and low temperature research in general. In particular, it is important to develop materials that are readily transferable to manufacturing, i.e., that can be used in combination with standard fabrication and interconnect techniques to reduce power dissipation and heating. Another critical metric is a relatively high critical temperature. With the advent of low-power, closed cycle cooling platforms, it is straightforward to obtain temperatures down to 2.2 K [6, 7], while the 4.2 K liquid-He barrier is more relevant to general-purpose low-temperature measurements. In either case, it is desired that the critical temperature be well above the base temperature in order to maintain a high critical current and avoid loss due to quasiparticles.

To that end, there is a small subset of existing superconducting materials that can be used for this application. In general, state-of-the-art materials include Nb and its binary and ternary alloys such as Nb-N, Nb-Ti and Nb-Ti-N. These materials all are useful in terms of high critical temperature, can be used in bulk or deposited as thin films, and can be connected using ultrasonic-wirebonding techniques. However, they tend to be difficult to work with mechanically and have poor soldering properties due to strong oxidation. While workarounds to these problems exist, they are not easily integrated into standard circuit fabrication.

Other typical materials, e.g. Pb, In, Sb, Al, Re, high T_c oxides, etc. suffer from toxicity, low melting temperatures, low critical temperatures, oxidation, or contact resistance. To date, no one solution has emerged that addresses all of the desired properties. One element in this latter category, Re, stands out as a promising candidate. Re is an transition metal that is resistant to oxidation with a high melting temperature of 3186 °C [8]. It is used widely in various industrial and scientific applications such as strengthening materials and in high-temperature thermocouples. While epitaxial Re can be used for low-loss rf-resonators in qubit circuits [9], it has no specific advantage over other more traditional materials. In particular, the critical temperature of crystalline-Re is relatively low, T_c(Re) $\approx$ 1.7 K [10].

However, T_c(Re) can be enhanced by strain, as pointed out in References 10, 11 and citations therein. This effect was attributed to an increase of the unit cell under shear strain. While the increased T_c(Re) does not go above the 4.2 K threshold mentioned above, it is similar to the effect seen in doping Re with W or Os [12]. Moreover, Chu points out in Reference 12 that there is a singularity close to the Fermi energy E_F that could drive anomalous behavior in the superconducting transition temperature through alloying. For example, it has recently been observed that Re forms a high T_c superconducting alloy with Mo [13, 14]. Exciting new results illustrate that this material has low rf loss, making it useful for resonators in quantum information applications [15]. Unfortunately, there is not a well known method of electroplating Mo-Re, making it challenging for general-purpose applications. There is, on the other hand, a body of work on electroplated Re films [16–21]; however, there has not yet been a systematic study of the superconducting properties of this system.

Here, we report on the superconducting nature of electroplated Re including electrical resistance, critical current, magnetization, and rf-loss. We prepare the Re in multilayers with Cu, Au, and Pd because carbonaceous

deposits tend to form on bare Re films. In addition, we are interested in evaluating the possible effects, i.e., strain and intermixing, of the different materials and interfaces on the superconducting properties. We study films grown on both Au/Si and Cu-clad circuit-boards. Type-II superconducting behavior is obtained with high critical-temperatures and critical-current densities and low rf-loss.

The Re was deposited using a dc power supply in the constant-current mode at 8 A/dm^2 from an aqueous solution containing 11 g/L of $KReO_4$ with the pH adjusted to 0.9 using sulfuric acid, according to the recipe in Reference 16. Deposition time was used to control layer thickness. The anodes were platinized titanium. Moderate stirring was done with a magnetic bar coated with polytetrafluoroethylene (PTFE) on a standard laboratory hot-plate stirrer at temperatures of 25-30 °C. Under these conditions the Re deposits were shiny and smooth, typically dark gray, with good adhesion. The various sample parameters are reported in Table I.

Sample \ Layer		Bottom	Middle	Top
1	Vacuum deposited trilayer on Si	200 Au	300 Re	100 Au
2	Electroplated bilayer on Si	200 Au	300 Re	n/a
3	Electroplated trilayer on Si	200 Au	300 Re	75 Au
4	Electroplated multilayer on Si	200 Au	(20 Au+75 Re)×10	75 Au
5	Electroplated multilayer on Cu/PTFE	200 Au	(20 Au+75 Re)×10	75 Au
6	Electroplated multilayer on Cu/PTFE	N/A	(500 Cu+75 Re)×5	500 Cu
7	Electroplated multilayer on Cu/PTFE	N/A	(500 Pd+75 Re)×5	500 Pd

TABLE I. Sample composition and thicknesses (nm). For sample 1, a quartz crystal thickness monitor was used. For sample 3 and 4, thicknesses were obtained directly from STEM. Samples 2, 5, 6 are nominal thicknesses based on times derived from 3 and 4. The Au, Cu, and Pd was grown using standard electroplating solutions with the exceptions being the Au metallization with 5 nm Ti adhesion layer on the Si in 2-4, and the vacuum prepared trilayers for 1.

To begin the study, we first grew a reference, Sample 1, of Au/Re/Au/prime-Si in vacuum using e-beam evaporation for the Au and sputter-deposition for the Re . This sample was used to compare against standard Re preparation methods.

Electroplated bilayers, trilayers, and multilayers of Re and Au where then grown on vacuum-prepared Au/Si substrates, Samples 2-4. A 20 nm seed of electroplated Au was typically applied before the Re.

Finally, multilayers of Re/Cu, Re/Au, and Re/Pd were electroplated directly onto commercial Cu/PTFE circuit boards, Samples 5-6. The boards were also patterned with resonators in order to test the rf properties of the multilayers. PTFE was used for its low loss tangent, tan $\delta = 6.8 \times 10^{-4}$, at 15 K [22].

Of particular concern in many low-temperature measurements is the dc-resistance and rf-loss that can create ohmic heating and degradation of quantum information. Therefore, we characterized both types of transport properties. The measurements were conducted in both a liquid-He dewar-based system and an adiabatic demagnetization refrigerator (ADR). The samples on Si were measured with a dc 4-point probe. For the films on the 35-μm-thick copper traces, it was necessary to use ac-transport with a lockin-amplifier due to their low resistance.

The resistivity of the sputtered-Re layers on Si was measured on ∼0.6 mm slices cleaved from 5-mm-wide strips. The T_c taken to be the first inflection point in the R vs. T curves. Samples were contacted by either wirebonding (25 μm Al wire for low current measurements) or soldering (standard Cu wire for high current measurements). All samples could be soldered using 60:40 SnPb. No problems were encountered wirebonding to the thick Cu while the for Pd- and Au-capped samples Al-wirebonds did not adhere as well.

Turning to the data, for the sputter-deposited Re samples we see a sharp transition to zero resistance in Figure 1(a)1, with T_c slightly exceeding that of the highly strained Re from Reference 10. This can be attributed to Au-Re interfacial strain that tends to expand the Re unit cell. Subsequent high temperature annealing, up to 400 °C, of this sample showed that T_c is stable within 0.1 K. This is in line with the high melting temperature of Re and its immiscibility with Au [24–26].

More surprisingly, for the electroplated samples on Si, a progression of T_c up to approximately 6 K was obtained. First, with the Re/Au bilayer sample in Figure 1(a)2, we measured a range of transition temperatures from T=4.3 K to 4.7 K. These samples tend to tarnish over a period of a few weeks due to the exposed Re surface. Therefore, we moved to capping the samples with electroplated Au films. This resulted in more stable films and reproducible T_c measurements. Moreover, T_c in these layered samples increased to T>5.7 K, with multiple steps in the transition in some samples, as shown in Figure 1(a)3. Experiments on these films, including sputtering the top Au film off and using much thicker Au layers, tend to depress the T_c. Subsequent samples grown as multilayers on Si, Figure 1(a)4, demonstrated even higher reproducibility and enhancement of T_c.

We then looked at the multilayered films on commercial circuit boards, Figure 1(a)5. Here, as T decreases we see the first drop in the resistance for $T \approx 6.1$ K. However, there are slight steps in these transitions with

low resistance tails. The tails go down an additional 0.1 – 0.5 K, depending on the current, before the transition to zero resistance.

In light of the variation of T_c(Re) due to strain, as outlined above, the properties of the Re grown with a series of metal films, i.e., Cu, Au, and Pd was compared. These metals all share a close-packed structure, fcc, that is similar to the hcp structure of Re. However, the nearest-neighbor distance in the close-packed plane for these three is 0.256, 0.275, and 0.288 nm [27], respectively. This spans compressive, low, and expansive strain relative to the 0.274 nm near-neighbor spacing for elemental Re[27]. As shown in Figure 1(b), all three types of multilayered films demonstrate the enhancement of T_c. Both Re/Au and Re/Cu have a sharp drop in resistance well above 6 K, whereas the Re/Pd samples have a slightly lower, more rounded transition. This may be due to the fact that Re, Cu, and Au are immiscible, while RePd tends to alloy[25].

The versatility in connecting to the samples, i.e. soldering and wirebonding, was important. This allowed us to achieve good thermalization while high current densities, J is applied through the soldered contacts. In the same measurement, voltage leads could be wirebonded, as shown in the inset of Figure 2. For samples on Si substrates we measured critical current $J_c \approx 2.5$ to 5×10^8 A/m^2 for trilayers and multilayers, respectively. For films on Cu/PTFE, data in Figure 2, we saw lower $J_c \sim 10^7$ A/m^2 due to the low temperature tails.

While it is significant that the resistance drops to zero, it does not unequivocally demonstrate that the films are superconducting over their entire area. Magnetic measurements, which access two dimensions, are able to distinguish this behavior. For this study, the Au/(AuRe)×10/Si multilayer film, Sample 4, was cleaved into a rectangular sample, 5.0 mm × 5.6 mm, for measurement in a magnetometer based on a superconducting quantum interference device (SQUID). In magnetic measurements of superconductors, the apparent magnetic moment is due to the field produced by shielding currents and trapped magnetic flux. The moment was measured in a perpendicular field of $\mu_0 H = 0.5$ mT as a function of increasing temperature after cooling in zero field (ZFC), and after cooling in a field (FC) of $\mu_0 H = 0.5$ mT, Figure 3(a). The critical temperature, $T_c \approx 5.4$ K, is the end point of the broad transition from the shielding state to the normal state seen in the ZFC curve. The positive FC moment values for $T < T_c$ is indicative of an incomplete Meissner effect (incomplete expulsion of magnetic flux upon cooling below T_c), which is characteristic of type-II superconductors [28].

A hysteresis loop of magnetic moment as a function of field was measured at $T = 1.8$ K, Figure 3(b). The shape of the loop is characteristic of type-II superconductors [28]. The magnetic moment approaches zero at an upper critical field $\mu_0 H_{c2} \approx 2.5$ T. Symmetrical flux jumps are evident in the descending branches of the hysteresis loop. The initial curve is in red, with a low-field detail in the inset that suggests small lower critical field $\mu_0 H_{c1} \approx 2$ mT.

For most superconducting applications, notably SQUID magnetometry and quantum computing, it is also important to establish that the films maintain low loss at rf into the GHz regime. Therefore, resonator loss tangent, tan $\delta = 1/Q_T$ vs. temperature was compared for Au/Re plated (sample 5) vs. bare copper traces. Here, Q_T is the total quality factor of the resonator. The samples were placed in a magnetically shieled environment and cooled to low temperature. The rf measurements were made using a vector network analyzer. The Q_T was determined using a a Lorenzian fit [29] to the resonance from a 18.2 mm × long grounded-coplanar-waveguide resonator on a 0.59 mm thick PTFE board. The width and gap of the resonatore were 0.81 and 0.076 mm, respectively. The coupling quality factor of the resonator to the feedline, $Q_C \sim 2 \times 10^4$. Because the Q_T was significantly lower than Q_C, the internal loss tan $\delta_i = 1/Q_i \approx 1/Q_T$, where Q_i is the internal quality factor. As shown in Figure 4, for the bare copper board tan $\delta_i \approx 2 \times 10^{-3}$ for $T < 8$ K. Below about $T = 250$ mK a slight increase appears. This increase is consistent with two-level system (TLS) loss [30], most likely in the PTFE.

For the Re/Au electroplated circuit board, on the other hand, a decrease for tan δ_i of $T < 5$ K is seen in Figure 4. This is expected in the case of a superconducting transition in the Re-Au plating. The loss drops to that of PTFE at low temperature [22], and the TLS increase appears again at very low temperature. This shows that the loss is limited by the material of the circuit board rather than the metal traces. Therefore, development of new, low-loss board materials is now an important consideration for these circuits. On the other hand, these data show that plating of metallic components will significantly improve their performance at rf as well as dc.

In conclusion, these films demonstrate a new superconducting phase of Re that is both interesting from a scientific point of view and technologically useful. The enhancement of T_c for electroplated-Re, independent of the intermediate metal indicates it is likely due to the electroplating process rather than interfacial or intermixing effects [23]. This points to bulk effects, e.g., growth-induced strain, hydrogen incorporation [19, 31], low density, or other morphology effects that may be at play. This is corroborated by preliminary STEM studies (not shown) that show high transmission through the Re layers with a striated pattern perpendicular to the plane. Our own density-functional-theory calculations confirm the immiscibility of Au/Re and point to the possibility of an increase in the density of states near E_F as the Re lattice expands, in agreement with Reference 12. These ongoing studies may help identify the relationship be-

tween morphology and superconductivity for Re.

From the practical perspective, a wide spectrum of applications can immediately benefit from this development. More specifically, these films can be integrated into a wide range of standard circuit, component, interconnect, cable, and shield fabrication processes.

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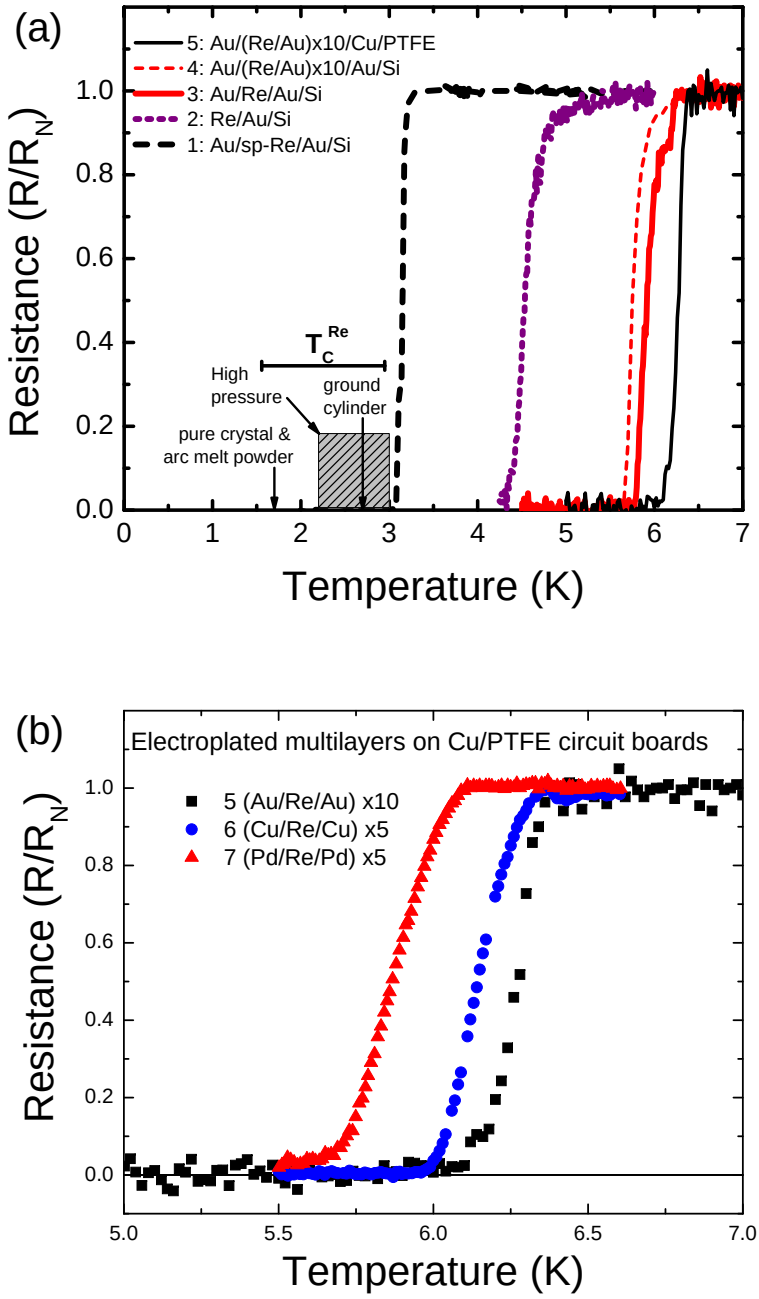

FIG. 1. Resistance vs. temperature. Panel (a) Data from References 10, 12 show T_c(Re) from the literature, curves show data from Au/Re samples described in Table I; Panel (b) comparison of critical behavior for Re in Cu, Au, and Pd multilayers, Samples 5-7.

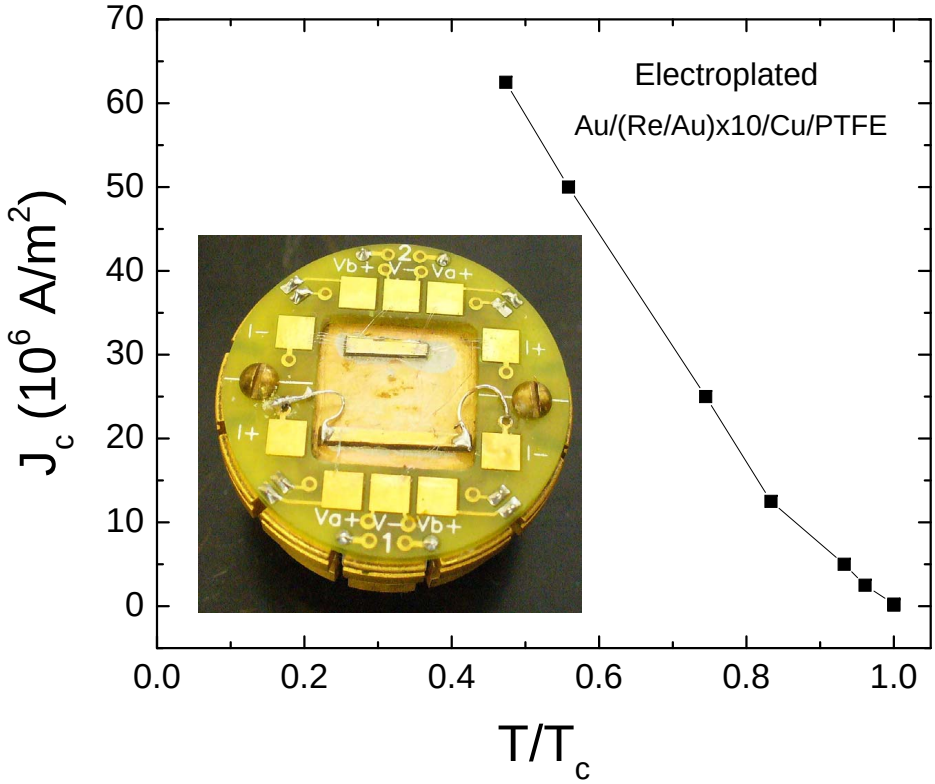

FIG. 2. Critical current J_c vs. normalized temperature $t = T/T_c$ for as-prepared electroplated multilayer, (Au/Re)x10/Au/PTFE, Sample 5. Inset shows two samples mounted; bottom sample with current leads soldered for high currents and voltage leads wirebonded; top sample is wirebonded on both current and voltage leads.

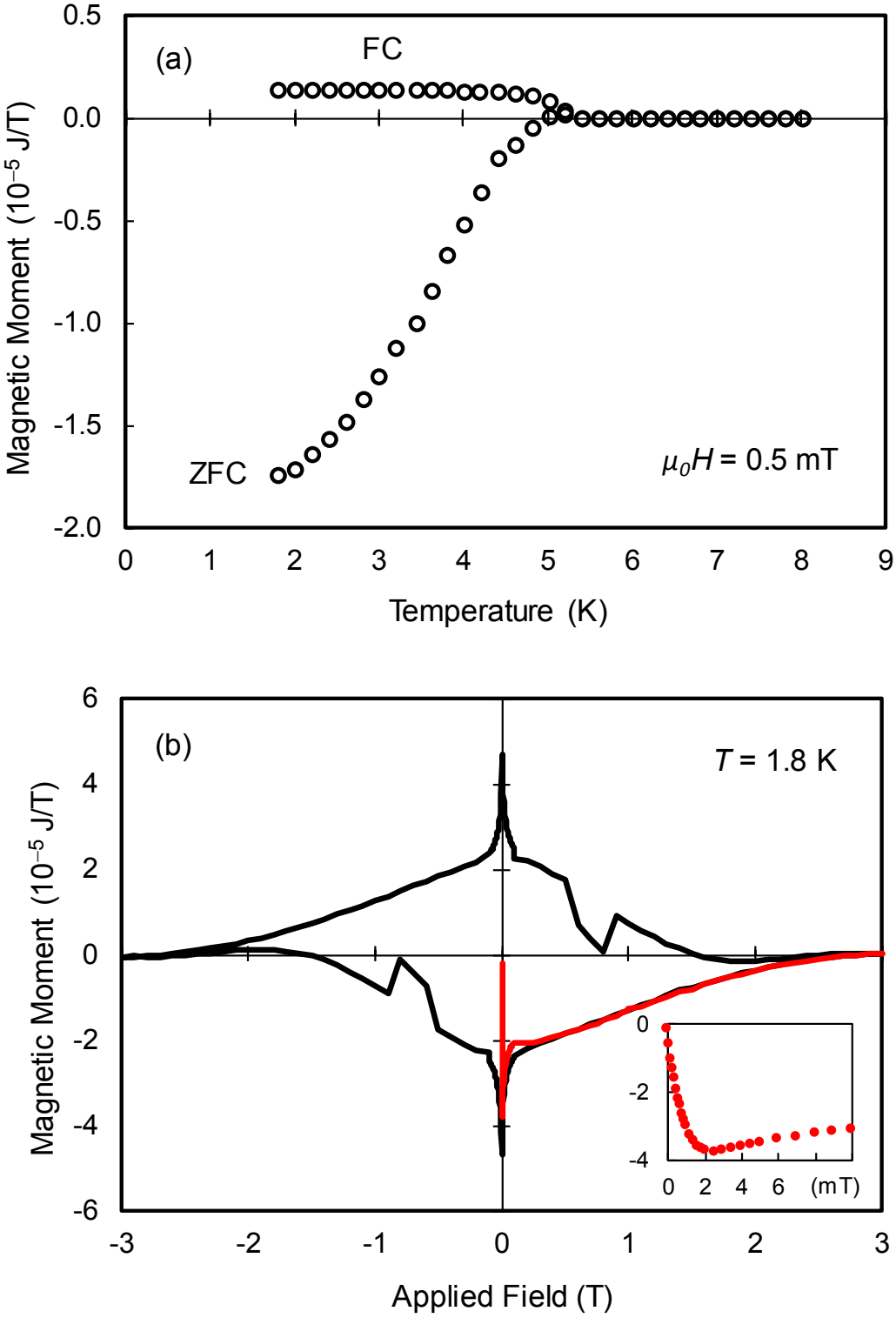

FIG. 3. Magnetic moment study of Sample 3, Panel (a) Zero-field cooled (ZFC) and field-cooled (FC) magnetic moment as functions of increasing temperature measured in $\mu_0 H = 0.5$ mT. The critical temperature is $T_c = 5.4$ K. Panel (b) Magnetic hysteresis loop measured at $T =$ 1.8 K. The initial curve with increasing field is in red. The upper critical field is $\mu_0 H_{c2} \approx 2.5$ T. Inset: Detail of the initial curve, which suggests a lower critical field of $\mu_0 H_{c1} \approx 2$ mT.

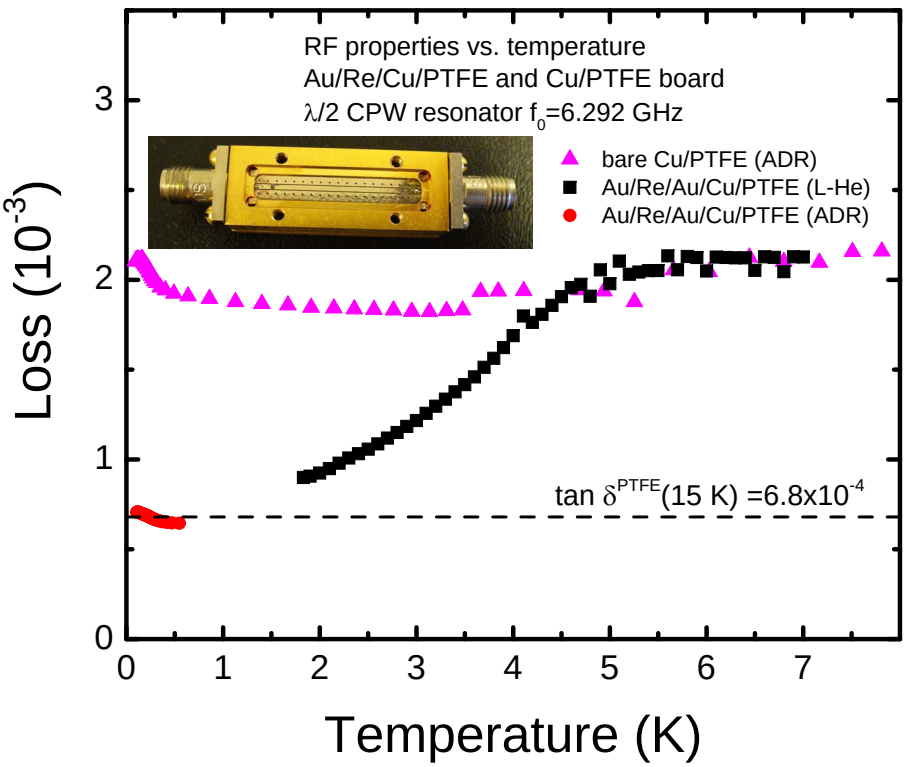

FIG. 4. Quality factor of Au/(Re/Au)x10/Cu/PTFE (Sample 5) compared to bare Cu/PTFE on a grounded coplanar resonator. The inset shows a Au/Re circuit board soldered into a Au-plated Cu box connectorized with SMA ports.